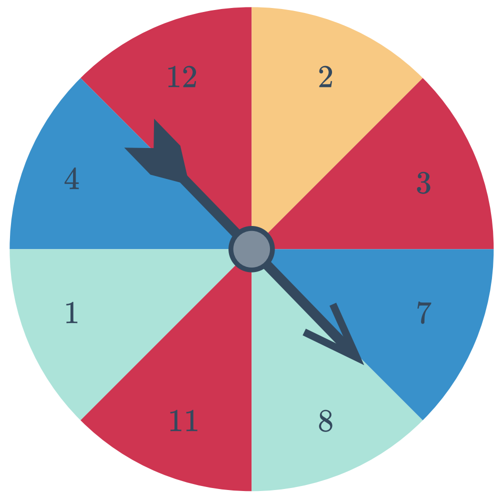
Language of probability

1 State the likelihood that corresponds to each of the following probability values using one of the following terms: Impossible, Unlikely, Likely or Certain

a $\frac{4}{5}$ b 0.1 c 1 d 0

e 0.6 f $\frac{9}{10}$ g 10% h $\frac{1}{8}$

2 Consider the spinner shown:



a State whether the following events have an even chance:

i Spinning an even number.

ii Spinning 7 or more.

iii Spinning 12 or less.

iv Spinning 7 or less.

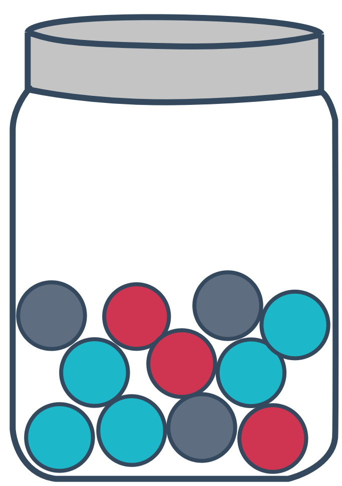
b State the likelihood of spinning 15 or more.

3 A standard six-sided die is rolled.

a State the likelihood that the outcome is a 2 or greater.

b State the likelihood that the outcome is a 7 or greater.

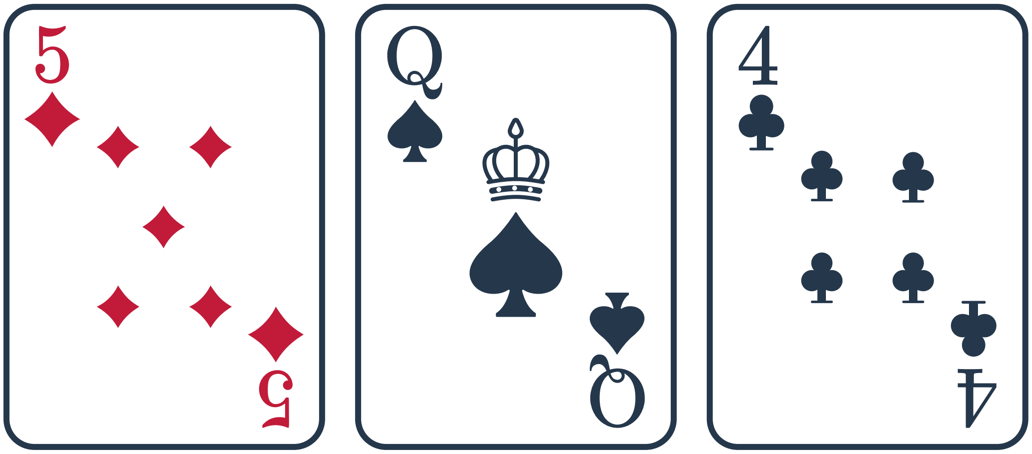
4 One marble is drawn out of the following jar containing 3 black, 3 red, and 6 blue marbles:



a State the color of the marble that you are most likely to draw.

b State the likelihood that you draw a marble that is not blue.

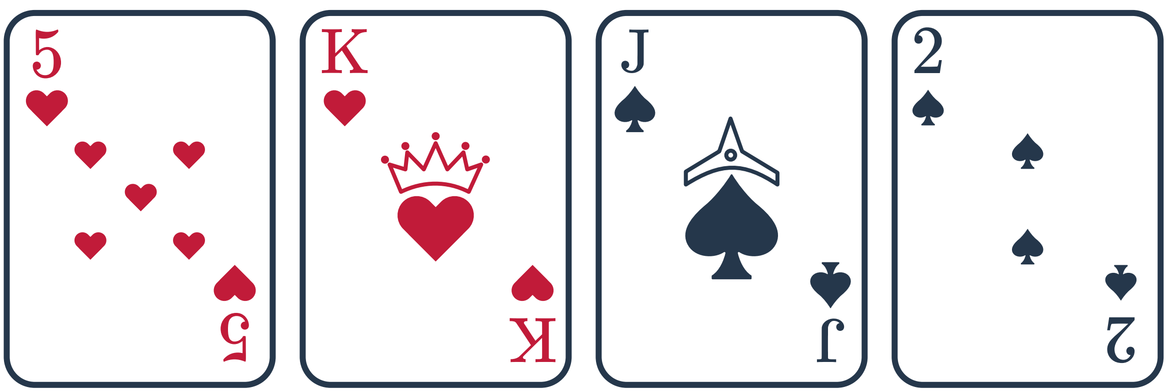
5 The three cards below are shuffled. One car[redacted]hen drawn at random:



a State the likelihood of drawing a Heart.

b State the likelihood of drawing a number card.

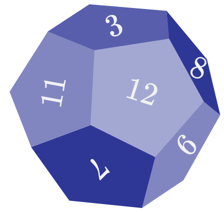
6 The four cards below are shuffled. One card is then drawn at random:



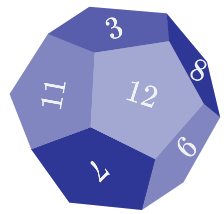
a State the likelihood of drawing a Jack, Queen or King.

b State the likelihood of drawing a card with a number less than 4.

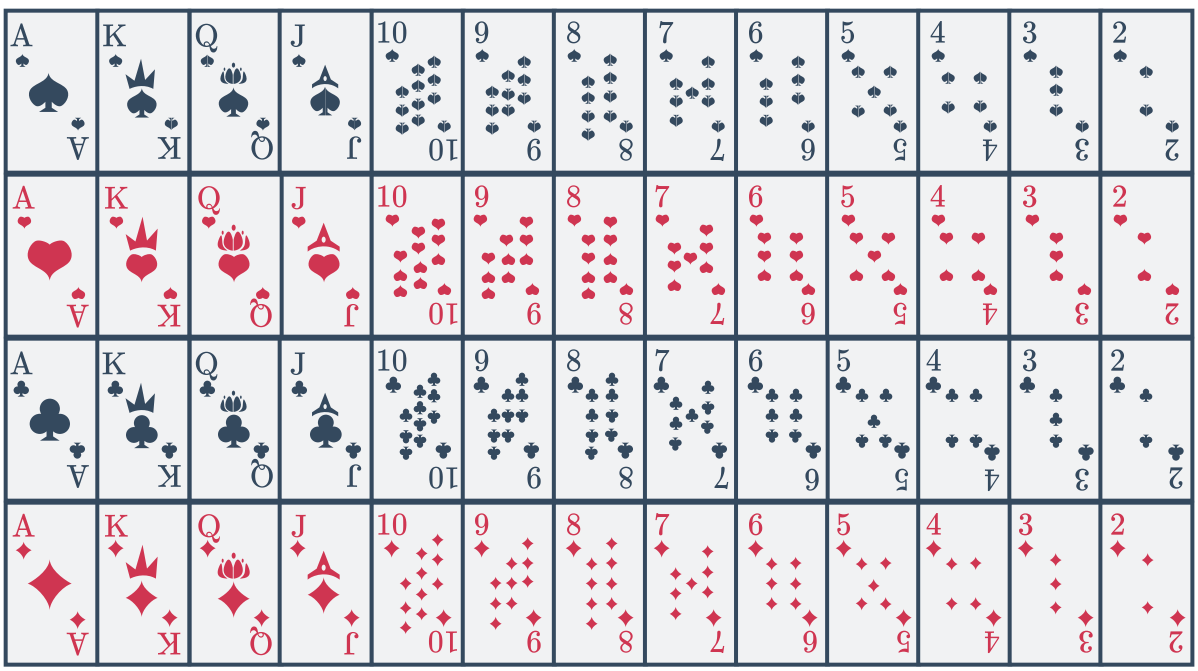
7 Kenneth has a six-sided die. Lisa has a twel[redacted]ed die. They decide to roll both their dice and the winner is the person who rolls the greatest. Who is more likely to win?



8 Vincent rolls a twelve-sided die and writes down whether he rolls 6 or more. If he does, he writes "yes", otherwize he writes "no". List all possible outcomes that correspond to the event where he writes "yes".



9 Rochelle draws a card from a standard 52-card deck, and writes down the suit of the card.



How many different outcomes could there be?

Theoretical probability

10 Estimate the probability of the following events as a decimal number:

a It will rain somewhere in America sometime this month.

b You meet someone today and correctly guess their birthday.

11 Which of these events has a probability closer to 0.25?

- You roll a six-sided die and get a 5 or more.
- You find $\$100$ on the ground today.

12 Which of these events has a probability closer to 0.75?

- You flip a coin ten times and get at least one tails.
- You roll a twelve-sided die and get 3 or more.

13 Given an event with a probability of 4%, determine whether the following statements are true or false:

a It is likely to occur.

b It is unlikely to occur.

c It should occur 4 times every 100 trials.

d It should occur 2 times every 100 trials.

e It should occur 20 times every 500 trials.

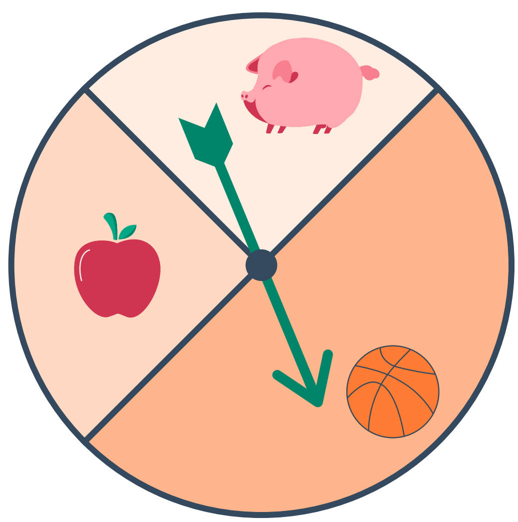
f It should never occur.

14 Consider the spinner shown.
Find the probability of spinning:

a An apple

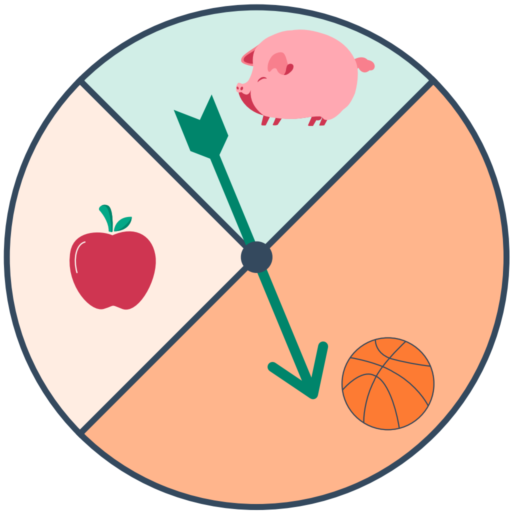
b A ball

c A pig

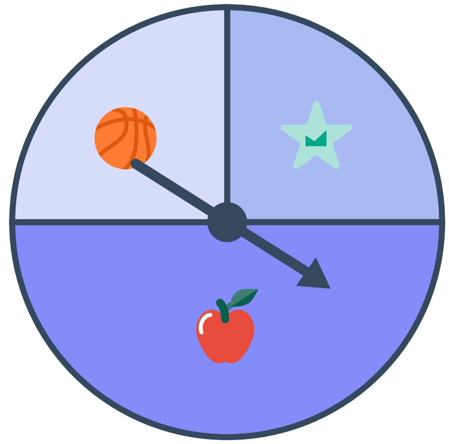


15 For each of the following spinners, find the [redacted]bility of getting a star. Write your answer as a percentage.

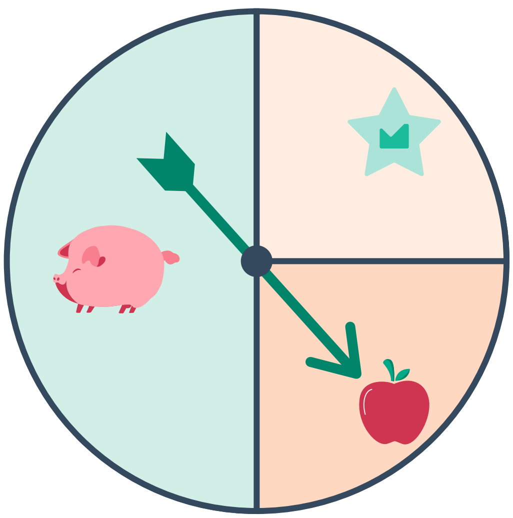
a



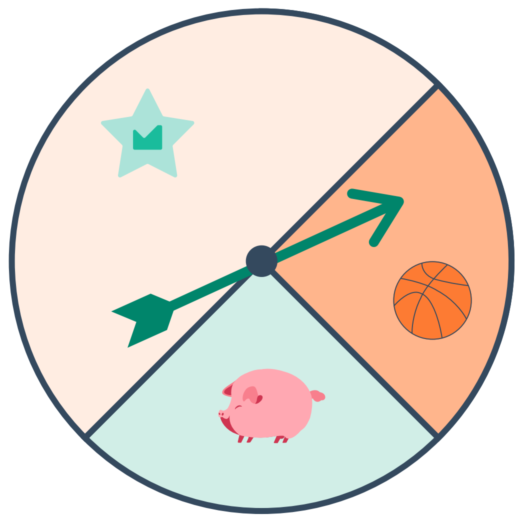
b



c

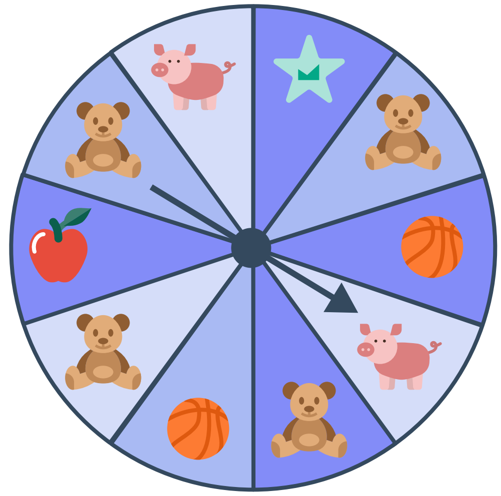


d



16 Paul has 23 marbles in a bag. 4 of them are black. Paul picks a marble from the bag without looking. Find the probability that Paul picks a black marble.

17 Consider the following spinner.
Find the probability of spinning:



a A pig.

b A ball or a pig.

c A pig or an apple.

d An apple or a star.

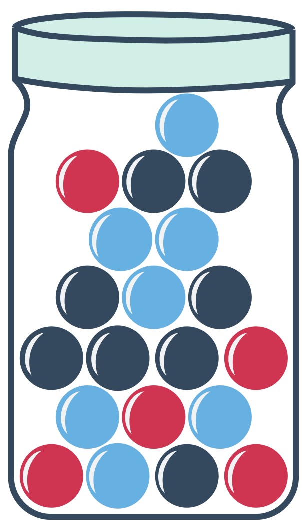
18 A spinner has ten sectors of the same size:

- 5 of the sectors show a star.
- 2 of the sectors show an apple.
- 3 of the sectors show an elephant.

Find the probability of spinning an elephant.

19 A bag contains 21 red marbles and 26 blue marbles. Find the probability of drawing a red marble.

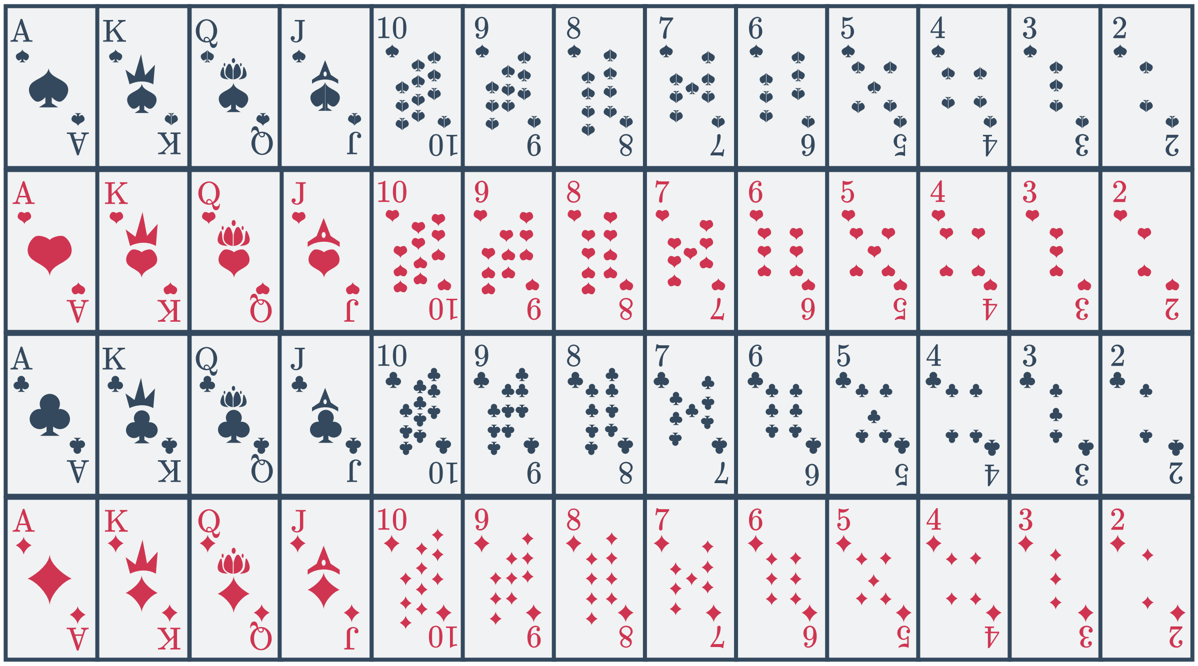
20 A single marble is drawn from the following jar which contains 5 red, 7 blue and 8 black marbles:



a Find the probability of drawing a blue marble. Write your answer as a percentage.

b Find the probability of drawing a blue marble or a red marble. Write your answer as a percentage.

21 A card is drawn from a standard deck of cards.



Find the probability that the card is:

a A diamond or a spade.

b A picture card (King, Queen or Jack).

c A spade or a red card.

d The Jack of Hearts or the Jack of Clubs.

22 A letter is chosen at random from the word COCOONS.

a Which letter is most likely to be chosen?

b Find the probability that the chosen letter is "C".

23 The numbers from 2 to 10 are written on separate cards. One card is chosen at random. Find the probability that the number is:

a A multiple of two.

b A prime number.

c A number less than 6.

24 A two-digit number is formed using the numbers 3 and 2. It can be two of the same or one of each number in any order.

a Find the probability that the number formed is odd.

b Find the probability that the number formed is more than 30.

25 Emma rolls two four-sided dice and multipli[redacted] two numbers she rolls together.

a Find the probability that the number formed is even.

b Find the probability that the number formed is larger than 6.

26 A full set of scrabble tiles is shown in the diagram below. The last two letters are the two "blank" tiles:

A	A	A	A	A	A	A	A	A	B
B	C	C	D	D	D	D	E	E	E
E	E	E	E	E	E	E	E	E	F
F	G	G	G	H	H	I	I	I	I
I	I	I	I	I	J	K	L	L	L
L	M	M	N	N	N	N	N	N	O
O	O	O	O	O	O	O	P	P	Q
R	R	R	R	R	R	S	S	S	S
T	T	T	T	T	T	U	U	U	U
V	V	W	W	X	Y	Y	Z		

One letter tile is drawn at random:

a Find the probability that it is a "G" or "R".

b Find the probability of drawing a vowel.

27 Luigi and Danielle enter a raffle where 80 tickets are on sale.

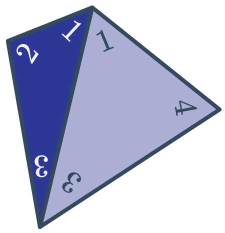
a Luigi buys 1 ticket. Find the probability that he wins if all the tickets are sold. Write your answer as a percentage.

b Danielle buys 5 tickets. Find the probability that she wins if all the tickets are sold. Write your answer as a percentage.

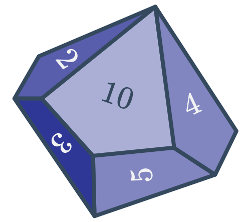
c By the end of the raffle 10 tickets were not sold. Find the probability that Danielle wins now. Write your answer as a percentage, and round it to two decimal places.

28 Consider the following dice:

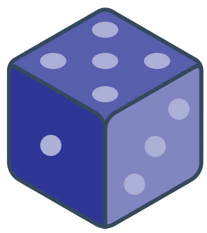
Four-sided die



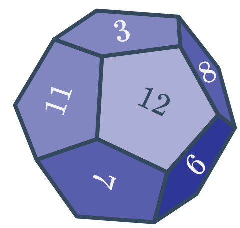
Ten-sided die



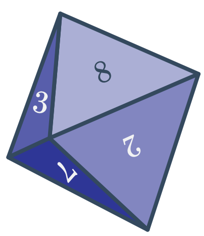
Six-sided die



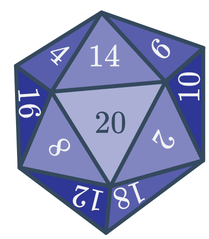
Twelve-sided die



Eight-sided die



Twenty-sided die



State the die that would give the greatest probability of rolling the following:

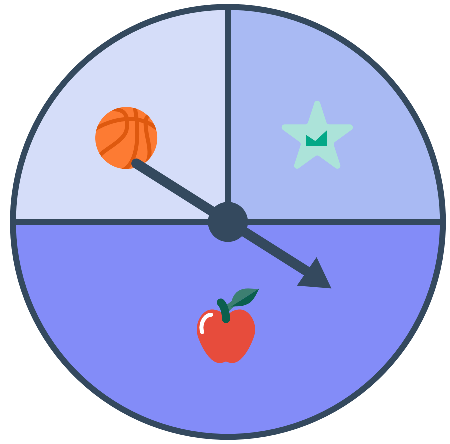
a 16 **b** 1 **c** 9

29 At a set of traffic lights, the green light is on for 24 seconds, then the yellow light lasts for 3 seconds, and then the red light is on for 13 seconds. This cycle then repeats continuously. If a car approaches the traffic light, calculate the probability that the light will be:

a Green **b** Yellow **c** Red

30 A six-sided die is rolled 24 times. How many times should we expect to roll a 1?

31 This spinner is spun 12 times.
Find the number of times we would expect to spin:



a An apple. **b** A ball.

32 If the probability of an event is $\frac{2}{3}$, how many times would you expect the event to occur in 18 trials?

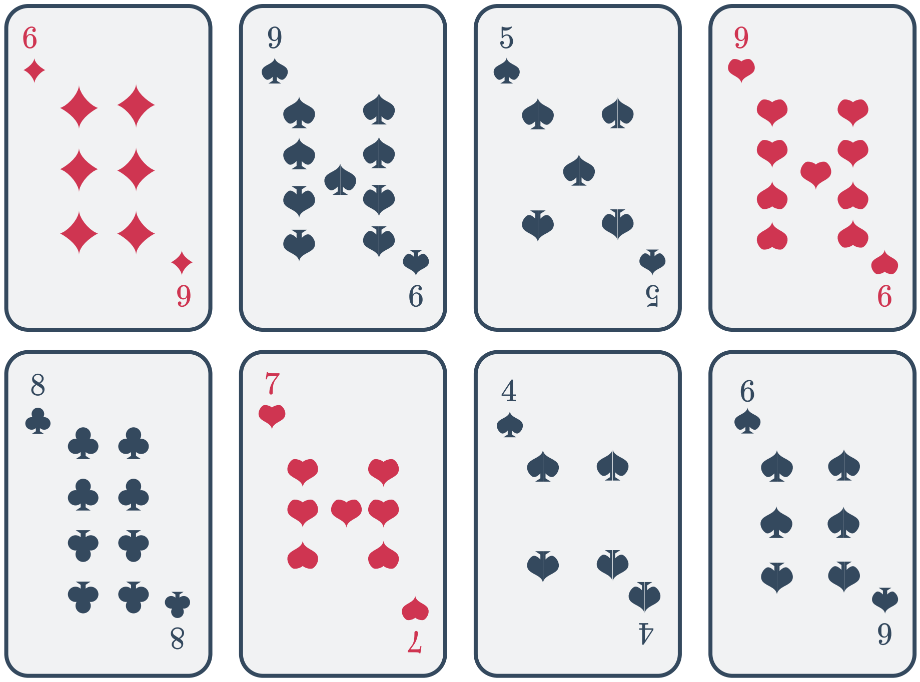
33 A twenty-sided die is rolled 100 times. How many times should we expect to roll a 14 or more? Round your answer to the nearest whole number.

34 A bag contains 28 red marbles, 27 blue marbles, and 26 black marbles.

a Find the probability of drawing a blue marble.

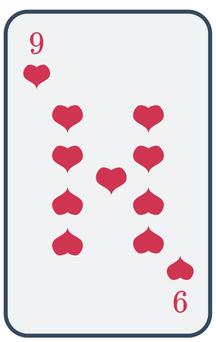
b A single trial is drawing a marble from the bag, writing down the color, and putting it back. If this trial is repeated 400 times, how many blue marbles should you expect? Round your answer to the nearest whole number.

35 Neil takes the following cards and shuffles t[redacted]up. He draws one card, shows it to you, and keeps it. You then draw one card from the remaining seven cards, and you win if the number on your card is greater than Neil's.

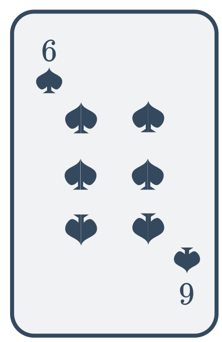


Find your chance of winning if Neil draws the following cards:

a



b



Experimental probability

36 An insurance company found that in the past year, of the 2558 claims made, 1493 of them were from drivers under the age of 25.

a Find the experimental probability that a claim is filed by someone under the age of 25, rounded to the nearest whole percent.

b Find the experimental probability that a claim is filed by someone 25 or older, rounded to the nearest whole percent.

37 The experimental probability that a commuter uses public transport is 50%. Out of 500 commuters, how many would you expect to use public transport?

38 A factory produces tablet computers. In March, it produced 8000 tablets, and 240 were found to be faulty.

a Calculate the experimental probability that a tablet produced by the factory is faulty.

b The factory plans to produce 9000 tablets in April. How many should they expect to be faulty?